Self-directed teaching system driven by video analysis feedback improves surgical skill training for clinical interns: a randomized controlled trial.

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Journal: BMC medical education | **Year:** 2026 | **DOI:** 10.1186/s12909-026-09104-y

Link: https://doi.org/10.1186/s12909-026-09104-y (open with institutional access)

Abstract

BACKGROUND: Clinical skills training in surgical clerkships relies heavily on faculty supervision, yet traditional models often face time constraints, lack of individualized guidance, and the inability to provide timely feedback. Technology-enhanced, learner-centred approaches, such as video-based self-directed systems, may help address these limitations. The Self-directed Teaching System (SDTS) integrates multi-angle video recording, self-paced practice, and structured feedback to support autonomous learning, but its effectiveness compared with that of conventional instructor-led teaching remains unclear. OBJECTIVE: This randomized controlled trial aimed to (1) evaluate the effectiveness and efficiency of SDTS versus traditional teaching in training basic surgical skills and (2) compare three SDTS feedback modalities (self-, peer-, and instructor-feedback) in influencing learning outcomes. METHODS: Sixty-two surgical interns were randomized to either a traditional teaching group (fixed 3-hour session with real-time faculty guidance) or an SDTS group (self-directed practice with multi-angle video recording and delayed feedback). Within the SDTS arm, participants were further randomized into self-feedback (n = 10), peer-feedback (n = 10), or instructor-feedback (n = 11) subgroups. All students completed pre- and post-training Objective Structured Clinical Examinations (OSCEs) on two skills: incision/suturing and suture removal/dressing change. Total training duration for the SDTS group, including video learning, practice, and feedback activities, was logged through the backend system data. RESULTS: Fifty-six interns completed the full protocol. For incision and suturing, SDTS shows no statistical difference compared with traditional teaching ($P > 0.05$). However, for suture removal/dressing change, a more complex, step-intensive procedure, the traditional group scored significantly higher than SDTS group ($P < 0.05$). Despite this, SDTS participants required substantially less total training time in both modules. The three SDTS feedback subgroups demonstrated no statistical differences in post-training OSCE performance. Learners reported high levels of satisfaction with the video-based feedback process and perceived improvement in procedural skills. CONCLUSIONS: The SDTS provides a flexible, technology-supported approach to surgical skills training and can deliver learning outcomes comparable to traditional methods for basic procedures while enabling more flexible use of training time. However, for more complex tasks, fully self-regulated practice without the required minimum training time may risk premature disengagement, before participants could fully refine their skills. While the feedback modality did not significantly impact outcomes, studies with a larger sample size are needed to verify subgroup effects. The SDTS shows strong potential as a scalable, learner-centred model for clinical education.

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